Experiment No. 6

Enumerate Web Server Information

AIM

To enumerate detailed information from a target web server such as supported HTTP methods, directory/file structure, installed modules, and hidden resources using enumeration tools.

PROCEDURE

1. Use Nmap's http-enum NSE script (nmap --script http-enum -p80) to enumerate common directories and files on the web server.
2. Use the OPTIONS HTTP method (curl -X OPTIONS -i) to list the HTTP methods supported by the server (GET, POST, PUT, DELETE, TRACE, etc.).
3. Use a directory brute-forcing tool such as DirBuster or Gobuster with a wordlist to discover hidden directories and files.
4. Use Nikto to perform an automated web server scan for outdated software, default files, and misconfigurations.
5. Enumerate server banner information and installed modules from response headers (e.g., mod_ssl, mod_php).
6. Record all discovered directories, files, supported methods, and potential misconfigurations.
7. Summarize findings highlighting any unnecessary HTTP methods enabled or sensitive files/directories exposed.

SAMPLE OUTPUT

```
$ nmap --script http-enum -p80 testserver.local
| http-enum:
|   /admin/:        Possible admin folder
|   /backup/:        Backup folder
|   /robots.txt:     Robots file
|   /wp-login.php:   WordPress login page

$ curl -X OPTIONS -i http://testserver.local
HTTP/1.1 200 OK
Allow: GET, POST, HEAD, OPTIONS, TRACE

Nikto scan summary:
  + Server leaks inodes via ETags
  + TRACE method is enabled (may allow XST attacks)
  + /backup/ directory listing enabled
```

RESULT

Detailed enumeration of the target web server was carried out, revealing hidden directories, supported HTTP methods (including the risky TRACE method), and server misconfigurations, using Nmap, Nikto, and manual HTTP probing.